RISHIKA VATSA

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Summary

Computer Science undergraduate (CGPA 8.81/10) with strong fundamentals in data structures, algorithms, and OOP (NPTEL Java, Top 5% nationally), and hands-on experience building REST APIs, SQL-backed services, and full-stack applications. Built an end-to-end Employee module in a Node.js/Supabase platform, contributed to a FastAPI backend at Smart India Hackathon 2025, and engineered a complete ML/IoT pipeline from hardware to web. Seeking **SDE / Backend Engineering** roles.

Experience

Full-Stack Developer Intern **Jun 2026 – Aug 2026**
EssProm Academy

- Built the Employee module end-to-end (frontend + backend) of a multi-user Resource/Project Management System in Node.js/Express and Supabase, with custom REST APIs for daily work logging, a centralized task view, and self-service timesheet submission.
- Contributed to role-based access control design and JWT/Supabase authentication middleware (`authenticateUser`, `requirePermission`) shared with the Manager module, securing a system that handles real user data.

Projects

Civic Issue Reporting App (Smart India Hackathon 2025) | *FastAPI, Supabase/PostgreSQL, REST APIs* **2025**

- Contributed in a 6-member team to a cross-platform prototype covering civic issue reporting through resolution tracking; implemented full CRUD operations on a FastAPI backend with ∼10 REST endpoints and a Supabase (PostgreSQL) database, enabling real-time sync and live status updates across devices.

EssProm Resource Management System | *Node.js, Express, Supabase, JWT* **Jun – Aug 2026**

- Built a project/task platform with Employee and Admin interfaces for assigning tasks, tracking progress, and collecting timesheets; owned the Employee module with custom REST APIs and session-secure JWT middleware.

Sign Language Interpreter Glove | *ESP32, Python, TensorFlow, IoT* **2025 – Present**

- Built a smart glove translating finger-sensor data into real-time speech: trained a TensorFlow gesture-classification model on cleaned sensor data, deployed it for edge inference on ESP32, and architected the full pipeline (sensors, ML model, backend API, web visualisation), iteratively reducing latency.

Holographic MIMO in Security (IDP Research Project) | *MATLAB, CST Studio Suite* **2024 – 2025**

- Analysed 10+ research papers on 6G physical-layer security and simulated beamforming models in MATLAB and CST Studio Suite, evaluating SNR and interception-risk metrics across antenna configurations.

Technical Skills

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST API Design, Schema Design
Languages: Python, Java, C++, C, SQL, JavaScript
Backend & Web: FastAPI, Node.js/Express, REST APIs, JWT/RBAC Auth, HTML, CSS, PHP
Databases: PostgreSQL, MySQL, MongoDB, Supabase
Data & Streaming: Apache Kafka (coursework), ETL/Data Pipeline Design, Data Cleaning
Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, XAMPP, Claude, ChatGPT, GitHub Copilot
ML & IoT: TensorFlow, NumPy, Pandas, OpenCV, SpeechRecognition, Neural Networks, ESP32, Arduino UNO

Education

B.Tech – Computer Science & Engineering (Core), Adamas University, Barasat **2023 – 2027**
CGPA: 8.81/10 (through Sem. 6)
Java & SQL Training Program, QSpider **2026**
Class XII (CBSE): 73.66% | **Class X (CBSE):** 93.88% | DAV Public School, Raniganj

Certifications & Achievements

NPTEL Java Programming: 88%, Top 5% nationally | **Forage:** Tata Group Data Analytics, Deloitte Australia Technology (Aug 2025) | **HackerRank:** 5-star C++